

Limitations of the Bohr Model.

The Bohr model was an important step in the development of atomic theory, however, it has several limitations:-

- ① It is in violation of the Heisenberg Uncertainty Principle. The Bohr model considers electrons to have both a known radius and orbit, which is impossible according to Heisenberg.
- ② The Bohr model is very limited in terms of size. For Spectral prediction one obtained when larger atoms are in question.
- ③ It can not predict the relative intensities of Spectral lines.
- ④ It does not explain the Zeeman Effect, when the Spectral line is split into several components in the presence of a magnetic field.
- ⑤ The Bohr model does not account for the fact that accelerating electrons do not emit electromagnetic radiation.

6

$$\begin{aligned}\frac{1}{\lambda} &= R \left(\frac{1}{5} - \frac{1}{\infty} \right) \text{ Since } \frac{1}{\infty} = 0 \\ &= R \left(\frac{1}{5} - 0 \right) \\ &= R/5 \\ \lambda &= 5/R = 5 / 1.097 \times 10^7 \\ &= 4.55788 \times 10^{-7} \text{ m.}\end{aligned}$$

6) The longest wavelength will be when $n_f = 5$ and $n_i = 6$

Hence, $\frac{1}{\lambda} = R \left(\frac{1}{5} - \frac{1}{6} \right)$

$$= R \left(\frac{1}{30} \right)$$

$$\begin{aligned}\lambda &= 30/R = 30 / 1.097 \times 10^7 \text{ m} \\ &= 2.734 \times 10^{-6} \text{ m}\end{aligned}$$

Note:

$E = \frac{1}{\lambda}$ hence, the bigger the atom, the smaller the wavelength of light or heat emitted.

3) An emission spectrum gives one of the lines in Balmer series of the hydrogen atom at 410 nm. This wavelength results from a transition from an upper energy level to $n=2$. What is the principal quantum number of the upper level.

$$\begin{aligned}\frac{1}{\lambda} &= R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \\ &= R \left(\frac{1}{2^2} - \frac{1}{n^2} \right) \\ \text{Since } R &= 1.097 \times 10^7 \text{ m}^{-1}, \lambda = 410 \text{ nm} \\ \frac{1}{410 \text{ nm}} &= 1.097 \times 10^7 \left[\frac{1}{2^2} - \frac{1}{n^2} \right] \\ n^2 &= \frac{-1}{-0.02778}\end{aligned}$$

$$\begin{aligned}36 &= n^2 \\ n &= 6\end{aligned}$$

The emission resulted from a transition from energy level 6 to energy level 2.

Assignment: 1

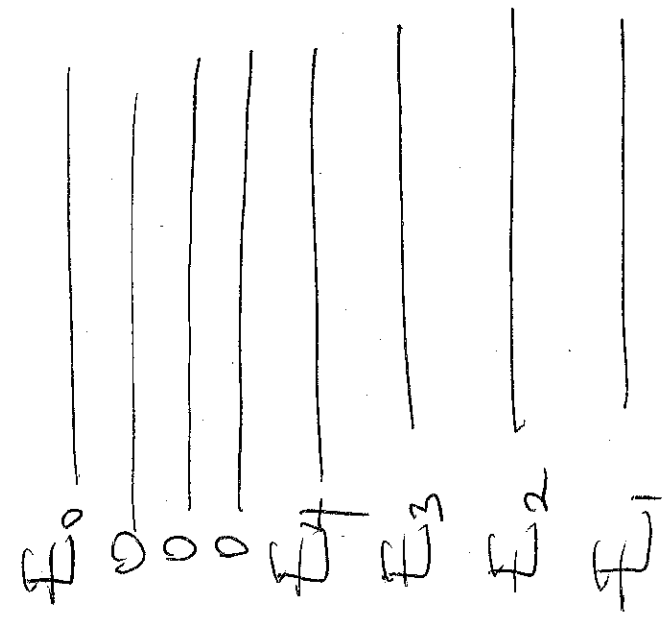
Derive an expression for the energy level of the Hydrogen Atom.

The Bohr model assumes that an electron bound to a nucleus in an atom can only move about the nucleus in certain orbits without radiating energy, but can make a transition from one level to the other, hence Bohr specified the following transition 'jumps' from one level to the other.

- ① Lyman Series: Transition to quantum level $n=1$ from higher level (i.e. $n_f = 1, n_i \geq 2$)
- ② Balmer Series: Transition to quantum level $n=2$ from higher levels (i.e. $n_f = 2, n_i \geq 3$)
- ③ Paschen Series: Transition to quantum level $n=3$ from higher levels (i.e. $n_f = 3, n_i \geq 4$)
- ④ Brackett Series: Transition to quantum level $n=4$ from higher levels (i.e. $n_f = 4, n_i \geq 5$)
- ⑤ Pfund Series: Transition to quantum level $n=5$ from higher levels (i.e. $n_f = 5, n_i \geq 6$)

These quantum jumps give rise to different radii and spectral lines as shown in the diagram below (overleaf)

(A)



Bohr Summarized his Postulates in four principles viz:

- ① Electrons assume only certain orbits around the nucleus. These orbits are stable and are called "Stationary Orbit".
- ② Each orbit has an energy associated with it. For example the orbit closest to the nucleus has an energy E_1 , the next E_2 and so on.
- ③ Light is emitted when an electron jumps from a higher orbit to a lower orbit to absorb when it jumps from a lower to a higher orbit.
- ④ The energy and frequency of light emitted or absorbed is given by the difference between the two orbits energies.

General Introduction:

A proper understanding of Materials helps the engineer avoid errors that has so directed the earlier ages of human existence -

Therefore, engineering material Science seeks to equip the engineers on the properties and structures of materials for efficient, effective and safe design. Understanding of how materials behave like and why they differ in properties now only possible with the atomistic understanding allowed by quantum mechanics.

Structure and Properties within the context of this study are:

Structure at the atomic level is the arrangement of atoms in different ways while properties are the material responds to the environment. For instance, the mechanical, electrical and magnetic properties are responses to mechanical, electrical and magnetic forces respectively.

In Summary, engineering material Science prepares the engineer in a world of variety of countless materials

① To be able to select a material for a given application based on considerations of cost and performance

② Be able to create a new material that will have some desirable properties

③ Understand the limits of materials and the change of their properties within a specific application.

Materials Science and Engineering.

Sometimes it is useful to subdivide the discipline of materials science and engineering into materials science and materials engineering subdiscipline. Strictly speaking, materials science involves investigating the relationship that exist between the structure and properties of materials (i.e. why materials have their properties). In contrast, materials engineering involves, on the basis of these structure-property correlation, designing or engineering the structure of a material to produce a predetermined set of properties. From a functional perspective, the role of materials scientist is to develop or synthesize new materials, whereas a materials engineer is called upon to create new products or systems using existing materials and/or to develop techniques for processing materials.

Most graduates in materials programs are trained to be both materials scientists and materials engineer. Structure is, at this point, a nebulous term that deserves some explanation. In brief, the structure of a material usually relates to the arrangement of its internal components.

Classification of Structural element on the basis of size

- ① Subatomic Structure: This involves electrons within the individual atoms, their energies and interactions with the nuclei.
- ② Atomic Structure: This relates to the organization of atoms to yield molecules or crystals.
- ③ Nanostructure: This deals with aggregates of atoms that form particles (nanoparticles) that have nanoscale dimensions (less than about 100 nm).

④ Microstructure: These are structural elements that are so small that one needs a microscope to observe them. Using some type of microscope (Structural feature having dimensions between 100nm and several millimeters)

⑤ Macrostructure: These are structural elements that may be viewed with the naked eye (with scale range between several millimetres and on the order of a meter).

The notion of property becomes elaboration. While in service use, all materials are exposed to external stimuli that evoke some types of responses. For example, a specimen subjected to forces experiences deformation, or a polished metal surface reflects light. A property is a material trait in terms of the kind and magnitude of response to a specific imposed stimulus. Generally, definition of properties are made independent of material shape and size.

Virtually all important properties of solid materials may be grouped into six different categories: mechanical, electrical, thermal, magnetic, optical, and deteriorative. For each, there is a characteristic type of stimulus capable of provoking different responses.

These are as follows:

① Mechanical Properties: Relate deformation to an applied load or force. Example include elastic modulus (Stiffness), strength, and resistance to fracture.

② Electrical Properties: The stimulus is an applied electric field, typical properties include electrical conductivity and dielectric constant

③ Thermal Properties: These relate to changes in temperature or temperature gradient across a material, examples of thermal behaviour include expansion and heat capacity

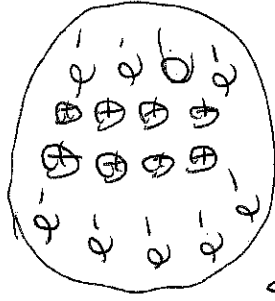
④ Magnetic Properties :- The responses of a material to the application of a magnetic field; Common magnetic properties include magnetic susceptibility and magnetization.

⑤ Optical Properties :- The stimulus is electromagnetic or light radiation; index of refraction and reflectivity are representative optical properties.

⑥ Deteriorative Properties :- Significant reduction in mechanical strength may result from exposure to elevated temperatures or corrosive environments.

Rutherford - Thomson Atom Model

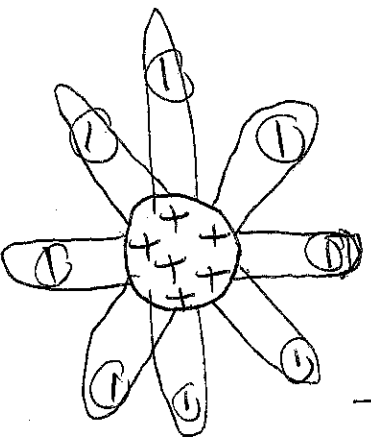
In the early 1900 the elements of the atom were already discovered to consist of subatomic particles called protons and electrons. However, it was not clear how these protons and electrons were arranged within the atom. J.J. Thomson suggested the "Plum pudding" model. In this model the electrons and protons are uniformly mixed throughout the atom.



J.J. Thomson Plum Pudding models of Atom.

In this model, electrons and protons are assumed to be in a state of equilibrium and even when exerted move or vibrate around this equilibrium position.

Rutherford a former student of Thomson in 1909 in a strange experience when imagined the atom as a "plum pudding" suggested the "Planetary Model" consisting of a positively charged relatively massive nucleus surrounded by electrons in a circularly motion. The finding came as part of an experience with X-rays he was shooting a beam of Alpha particles (or alpha rays, emitted by the radioactive element Radium) at a sheet of gold foil only $1/3000$ of an inch thick and tracing the particles paths. Most of the particles went right through the foil which would be expected if the atoms in the gold were like the plum pudding. But every now and then, a particle bounced back as though it had hit something solid. After tracing many particles and saw that majority of the alpha particles were deflected through small angles while very small number (about $1/5000$) were deflected through angles larger than 90° and up to 180° . Rutherford reason that the way the alpha particles could be deflected backwards was as if most of the mass and positive charge in an atom was concentrated in a nucleus about $10,000$ times smaller than the atom itself. All of the negative charges was held in electrons which must orbit the dense nucleus like planets around the Sun.



Bohr's Hydrogen Atom

Niels Bohr introduced the atomic Hydrogen model in 1913. He described it as a positively charged nucleus, comprised of protons and neutrons, surrounded by a negatively charged electron cloud. In the model, electrons orbit the nucleus in atomic shells. The atom is held together by electrostatic forces between the positive nucleus and negative surrounding. Bohr used quantum theory to explain the stability of most atoms. He found that the ratio of energy in electrons and the frequency of their orbit around their orbits around the nucleus was equal to Planck's Constant. Bohr then suggested the revolutionary idea that electrons "jump" between energy levels (orbit) in a quantum fashion; that is without ever existing in an in-between state. Thus when an atom absorbs or gives off energy (either as light or heat), the electron jumps to higher or lower orbits. According to Bohr, when an atom is excited by the input of energy, an electron moves to an orbit of greater radius, subsequently, when the electron moves back to the ground of energy, energy will be given off or released at some specific frequency given rise to a "Spectral line".

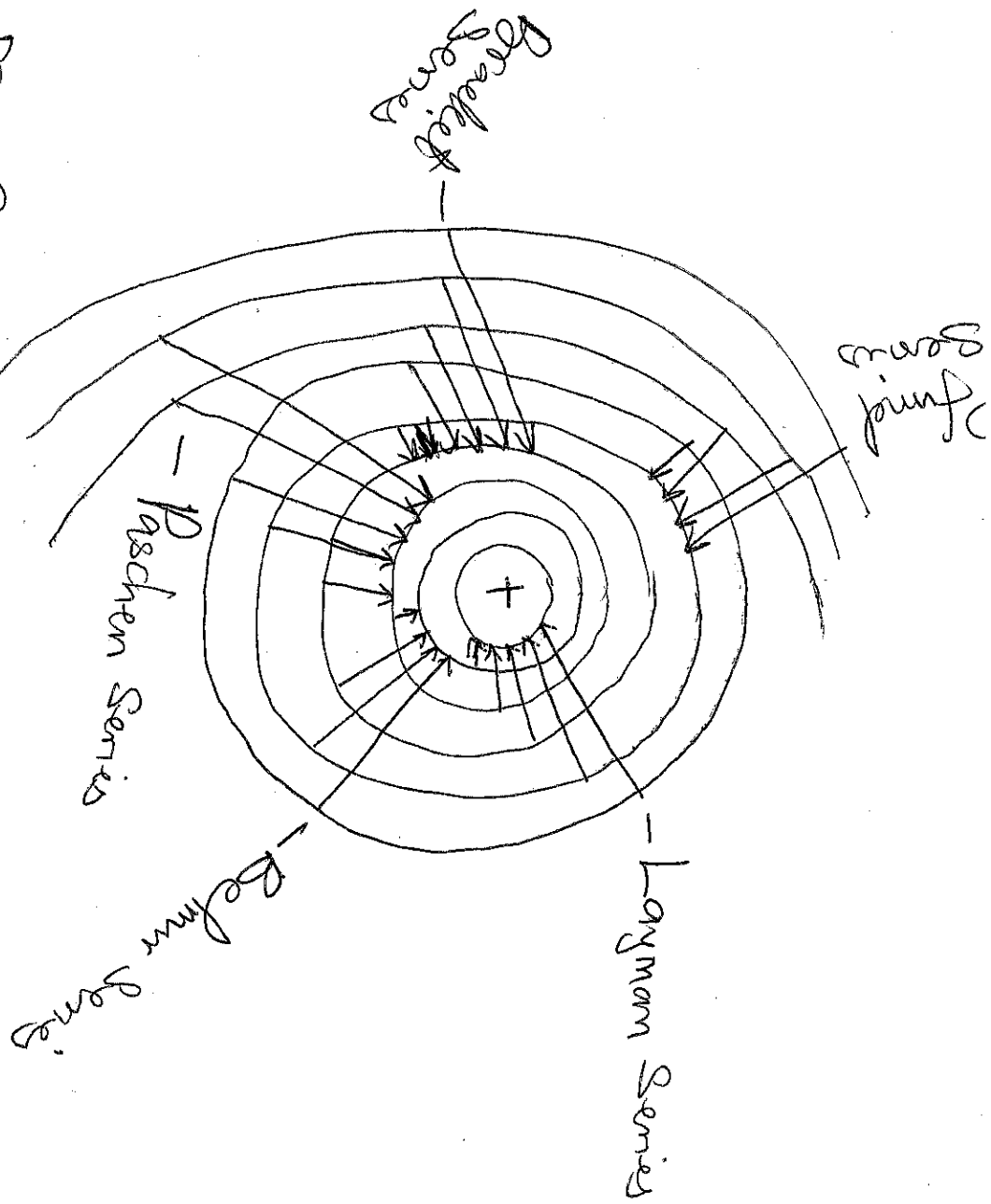


Fig:- Circular Bohr Orbits for the Hydrogen atom showing the transition which result in the different spectral lines.

Assignment : Explain the following Phenomena in terms of Bohr's theory.

- Spectral lines.
- Photo-Electric Emission.
- Electron Transmission from One Energy to Another.

Worked Example:

- ① Determine the wavelength of the photon emitted in the process of excitation of an electron falling from an energy -19.6 eV to ground level of energy -32.8 eV

Solution

$$\begin{aligned} E &= E_f - E_i \\ &= E_{\text{ground}} - E_{\text{excited}} \\ &= (32.8 - 19.6) \text{ eV} \\ &= (32.8 - 19.6) (1.60219 \times 10^{-19}) \text{ J} \\ E &= hf = hc/\lambda \end{aligned}$$

but $c = 3 \times 10^8 \text{ m/s}$

$$h = 6.6 \times 10^{-34} \text{ Js}$$

$$\begin{aligned} \therefore \lambda &= hc/E = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{2.115 \times 10^{-18}} \\ &= \underline{\underline{9.3622 \times 10^{-8} \text{ m}}} \end{aligned}$$

- ② Calculate the shortest wavelength in the Pfund series of the hydrogen spectrum given that Rydberg's Constant $R = 1.09 \times 10^7 \text{ m}^{-1}$, also calculate the longest wavelength.

Solution

In the Pfund series, the shortest wavelength occurs $n_f = 5$ and $n_i > \infty$

Using the formula

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$